

Technical Data Sheet: KNSB Propellant Grain Fabrication Log (Batch 03)

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Abstract

This technical document records the empirical manufacturing logs, thermal profiles, and dehydration kinetics for an experimental fuel-rich Potassium Nitrate and Sorbitol (KNSB) solid propellant grain processed on Wednesday, 8th July 2026. Departing from the standard 65:35 composition baseline, this batch investigates a modified 60:40 mass ratio. The primary research objective is to successfully decrease the motor's thrust profile without compromising the essential post-cure mechanical and structural properties of the grain matrix.

Dehydration operations commenced at 10:13 AM with an initial fuel charge of 866.0 g of liquid Sorbitol. The mixture reached its operational saturation point at 619.0 g after 80 minutes of monitored heating, at which point 929.0 g of Potassium Nitrate was integrated to satisfy the altered stoichiometric ratio.

Section 1: Batch Specifications & Material Metrics

The formulated mass breakdowns, core ratio adjustments, and casing parameters for Batch 03 are established in Table I.

Table I: Fuel-Rich Batch Specifications

Physical Parameter / Specification	Target / Achieved Value
Experimental Assignment	Thrust Profile Suppression Study
Target Propellant Formulation	60:40 KNSB (KN:Sorbitol)
Initial Fuel Mass (Sorbitol 70)	866.0 g
Theoretical Anhydrous Target	606.0 g
Achieved Processed Fuel Mass	619.0 g (Operational Saturation Point)
Measured Oxidizer Mass KN	929.0 g

Liner Casing Tare Weight	100.0 g
Total Dry Slurry Net Yield	1548.0 g